

CS 323: Final Project - Legend

< prog >	:=	P
< identifier >	:=	I
< dec-list >	:=	B
< dec >	:=	C
< type >	:=	K
< stat-list >	:=	G
< stat >	:=	H
< write >	:=	W
< str >	:=	J
< assign >	:=	A
< expr >	:=	E
< term >	:=	T
< factor >	:=	F
< number >	:=	N
< sign >	:=	S
< digit >	:=	D
< letter >	:=	L
< expr >'	:=	Q
< term >'	:=	R
intermediate 1	:=	X
intermediate 2	:=	Y
★ Collision solve 1	:=	Z
★ collision solve 2	:=	V

Terminals

program ; var begin end λ : , integer "value=",

= + - * / () print 0 1 2 3 4

5 6 7 8 9 a b c d l f

CS 323: Final Project - Removing EBNF, Left Recursion, and Collisions for Table

EBNF

$$I \rightarrow L \{L|D\}$$

X

$$N \rightarrow SD \{D\}$$

Y

$$\begin{aligned} I &\rightarrow LX \\ X &\rightarrow LX \\ X &\rightarrow DX \\ X &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} N &\rightarrow SDY \\ Y &\rightarrow DY \\ Y &\rightarrow \lambda \end{aligned}$$

Left Recursion

$$\begin{aligned} E &\rightarrow E + T^\alpha \\ E &\rightarrow E - T^\alpha \\ E &\rightarrow T/\beta \end{aligned}$$

$$\begin{aligned} E &\rightarrow TQ \\ Q &\rightarrow +TQ \\ Q &\rightarrow -TQ \\ Q &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} T &\rightarrow T * F^\alpha \\ T &\rightarrow T / F^\alpha \\ T &\rightarrow F/\beta \end{aligned}$$

$$\begin{aligned} T &\rightarrow FR \\ R &\rightarrow *FR \\ R &\rightarrow /FR \\ R &\rightarrow \lambda \end{aligned}$$

Collision Resolution

- The grammar below cause issues when building the predictive parsing table. These grammar create two options in certain table cells which will not work for parsing. Below is the attempt to resolve these issues.

$$\begin{aligned} C &\rightarrow I, C \\ C &\rightarrow I, Z \end{aligned}$$

$$\begin{aligned} \Rightarrow C &\rightarrow IZ \\ Z &\rightarrow ,C \\ Z &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} G &\rightarrow H, G \\ G &\rightarrow H, V \end{aligned}$$

$$\begin{aligned} \Rightarrow G &\rightarrow HV \\ V &\rightarrow G \\ V &\rightarrow \lambda \end{aligned}$$

$P \rightarrow \text{program } I ; \text{ var } B \text{ begin } G \text{ end}$

$I \rightarrow L X$

$X \rightarrow L X$

$X \rightarrow D X$

$X \rightarrow \lambda$

$B \rightarrow C : K ;$

* $C \rightarrow I Z$

* $Z \rightarrow , C$

* $Z \rightarrow \lambda$

$K \rightarrow \text{integer}$

* $G \rightarrow H V$

* $V \rightarrow G$

* $V \rightarrow \lambda$

$H \rightarrow W$

$H \rightarrow A$

$W \rightarrow \text{print}(J I)$

$J \rightarrow \text{"value="},$

$J \rightarrow \lambda$

$A \rightarrow I = E ;$

$E \rightarrow T Q$

$Q \rightarrow + T Q$

$Q \rightarrow - T Q$

$Q \rightarrow \lambda$

$T \rightarrow F R$

$R \rightarrow * F R$

$R \rightarrow / F R$

$R \rightarrow \lambda$

$F \rightarrow I$

$F \rightarrow N$

$F \rightarrow (E)$

$N \rightarrow S D Y$

$Y \rightarrow D Y$

$Y \rightarrow \lambda$

$S \rightarrow +$

$S \rightarrow -$

$S \rightarrow \lambda$

$D \rightarrow 0$

$D \rightarrow 1$

$D \rightarrow 2$

$D \rightarrow 3$

$D \rightarrow 4$

$D \rightarrow 5$

$D \rightarrow 6$

$D \rightarrow 7$

$D \rightarrow 8$

$D \rightarrow 9$

$\downarrow$

$L \rightarrow a$

$L \rightarrow b$

$L \rightarrow c$

$L \rightarrow d$

$L \rightarrow l$

$L \rightarrow f$

$$\text{FIRST}(L) = \{a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(D) = \{0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(S) = \{+ \ - \ \lambda\}$$

$$\text{FIRST}(Y) = \{\lambda \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(N) = \{+ \ - \ \lambda \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(F) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(R) = \{* \ / \ \lambda\}$$

$$\text{FIRST}(T) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(Q) = \{+ \ - \ \lambda\}$$

$$\text{FIRST}(E) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(A) = \{a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(J) = \{\lambda \ \text{"value="} \}$$

$$\text{FIRST}(W) = \{\text{print}\}$$

$$\text{FIRST}(H) = \{\text{print} \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(V) = \{\lambda \ \text{print} \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(\overset{G}{\#}) = \{\text{print} \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(k) = \{\text{integer}\}$$

$$\text{FIRST}(Z) = \{\lambda \ , \}$$

$$\text{FIRST}(C) = \{a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(B) = \{a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(X) = \{\lambda \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(I) = \{a \ b \ c \ d \ l \ f\}$$

$$\text{FIRST}(P) = \{\text{program}\}$$

$$\text{FOLLOW}(P) = \{ \$ \}$$

P is starting

$$\therefore \$ \in \text{FOLLOW}(P)$$

$$\text{FOLLOW}(I) = \{ ; , \text{ } : \} = \{ \}$$

P $\rightarrow$ program I ; var B begin G end

$$\therefore ; \in \text{FOLLOW}(I)$$

C $\rightarrow$ I Z

$$\therefore \text{FIRST}(Z) \in \text{FOLLOW}(I)$$

Z $\rightarrow$ $\lambda \Rightarrow$ C $\rightarrow$ I

$$\therefore \text{FOLLOW}(C) \in \text{FOLLOW}(I)$$

W $\rightarrow$ print (JI)

$$\therefore) \in \text{FOLLOW}(I)$$

A $\rightarrow$ I = E ;

$$\therefore = \in \text{FOLLOW}(I)$$

$$\text{FOLLOW}(X) = \{ ; , : \} = \{ \}$$

I $\rightarrow$ L X

$$\therefore \text{FOLLOW}(I) \subseteq \text{FOLLOW}(X)$$

X $\rightarrow$ L X

no new info

X $\rightarrow$ D X

no new info

X $\rightarrow$ λ

no new info

$$\text{FOLLOW}(B) = \{ \text{begin} \}$$

P $\rightarrow$ program I ; var B begin G end

$$\therefore \text{begin} \in \text{FOLLOW}(B)$$

$$\text{FOLLOW}(C) = \{ : \}$$

B $\rightarrow$ C : K ;

$$\therefore : \in \text{FOLLOW}(C)$$

Z $\rightarrow$, C

$$\therefore \text{FOLLOW}(Z) \subseteq \text{FOLLOW}(C)$$

$$\text{FOLLOW}(Z) = \{ : \}$$

C $\rightarrow$ I Z

$$\therefore \text{FOLLOW}(C) \subseteq \text{FOLLOW}(Z)$$

$$\text{FOLLOW}(K) = \{ ; \}$$

B $\rightarrow$ C : K ;

$$\therefore ; \in \text{FOLLOW}(K)$$

$$\text{FOLLOW}(G) = \{ \text{end} \}$$

P $\rightarrow$ program I ; var B begin G end

$$\therefore \text{end} \in \text{FOLLOW}(G)$$

V $\rightarrow$ G

$$\therefore \text{FOLLOW}(V) \subseteq \text{FOLLOW}(G)$$

$$\text{FOLLOW}(V) = \{ \text{end} \}$$

G $\rightarrow$ H V

$$\therefore \text{FOLLOW}(G) \subseteq \text{FOLLOW}(V)$$

$$\text{FOLLOW}(H) = \{ \text{print a b c d l f } \text{ } \text{end} \}$$

G $\rightarrow$ H V

$$\therefore \text{FIRST}(V) \in \text{FOLLOW}(H)$$

V $\rightarrow$ λ G $\rightarrow$ H

$$\therefore \text{FOLLOW}(G) \in \text{FOLLOW}(H)$$

$$\text{FOLLOW}(W) = \{ \text{print a b c d l f end} \}$$

H $\rightarrow$ W

$$\therefore \text{FOLLOW}(H) \subseteq \text{FOLLOW}(W)$$

$$\text{FOLLOW}(J) = \{ \text{a b c d l f} \}$$

W $\rightarrow$ print (JI)

$$\therefore \text{FIRST}(I) \in \text{FOLLOW}(J)$$

$$\text{FOLLOW}(A) = \{ \text{print a b c d l f end} \}$$

H $\rightarrow$ A

$$\therefore \text{FOLLOW}(H) \subseteq \text{FOLLOW}(A)$$

$$\text{FOLLOW}(E) = \{ ; \}$$

$$A \rightarrow I = E ;$$

$$F \rightarrow (E)$$

$$\therefore ; \in \text{FOLLOW}(E)$$

$$\therefore) \in \text{FOLLOW}(E)$$

$$\text{FOLLOW}(Q) = \{ ; \}$$

$$E \rightarrow TQ$$

$$Q \rightarrow +TQ$$

$$Q \rightarrow -TQ$$

$$Q \rightarrow \lambda$$

$$\therefore \text{FOLLOW}(E) \subseteq \text{FOLLOW}(Q)$$

no new info

no new info

no new info

$$\text{FOLLOW}(T) = \{ + - \neq ; \}$$

$$E \rightarrow TQ$$

$$Q \rightarrow \lambda \Rightarrow E \rightarrow T$$

$$\left. \begin{array}{l} Q \rightarrow +TQ \\ Q \rightarrow \lambda \end{array} \right\} Q \rightarrow +T$$

$$Q \rightarrow -TQ$$

$$\therefore \text{FIRST}(Q) \subseteq \text{FOLLOW}(T)$$

$$\therefore \text{FOLLOW}(E) \subseteq \text{FOLLOW}(T)$$

$$\therefore \text{FOLLOW}(Q) \subseteq \text{FOLLOW}(T)$$

no new info

$$\text{FOLLOW}(R) = \{ + - ; \}$$

$$T \rightarrow FR$$

$$R \rightarrow *FR$$

$$R \rightarrow /FR$$

$$R \rightarrow \lambda$$

$$\therefore \text{FOLLOW}(T) \subseteq \text{FOLLOW}(R)$$

no new info

no new info

no new info

$$\text{FOLLOW}(F) = \{ * / \neq + - ; \}$$

$$T \rightarrow FR$$

$$R \rightarrow \lambda \Rightarrow T \rightarrow F$$

$$R \rightarrow *FR \Rightarrow R \rightarrow *F$$

$$R \rightarrow \lambda$$

$$R \rightarrow /FR$$

$$\therefore \text{FIRST}(R) \subseteq \text{FOLLOW}(F)$$

$$\therefore \text{FOLLOW}(T) \subseteq \text{FOLLOW}(F)$$

$$\therefore \text{FOLLOW}(R) \subseteq \text{FOLLOW}(F)$$

no new info

$$\text{FOLLOW}(N) = \{ * / + - ; \}$$

$$F \rightarrow N$$

$$\therefore \text{FOLLOW}(F) \subseteq \text{FOLLOW}(N)$$

$$\text{FOLLOW}(Y) = \{ * / + - ; \}$$

$$N \rightarrow SDY$$

$$Y \rightarrow DY$$

$$Y \rightarrow \lambda$$

$$\therefore \text{FOLLOW}(N) \subseteq \text{FOLLOW}(Y)$$

no new info

no new info

$$\text{FOLLOW}(S) = \{ 0 1 2 3 4 5 6 7 8 9 \}$$

$$N \rightarrow SDY$$

$$\therefore \text{FIRST}(D) \subseteq \text{FOLLOW}(S)$$

$$\text{FOLLOW}(D) = \{ 0 1 2 3 4 5 6 7 8 9 \neq * / + - ; \}$$

$$N \rightarrow SDY$$

$$Y \rightarrow \lambda \Rightarrow N \rightarrow SD$$

$$X \rightarrow DX \Rightarrow X \rightarrow D$$

$$X \rightarrow \lambda$$

$$N \rightarrow SDY \Rightarrow N \rightarrow D$$

$$Y \rightarrow \lambda$$

$$S \rightarrow \lambda$$

$$\therefore \text{FIRST}(Y) \subseteq \text{FOLLOW}(D)$$

$$\therefore \text{FOLLOW}(N) \subseteq \text{FOLLOW}(D)$$

$$\therefore \text{FOLLOW}(X) \subseteq \text{FOLLOW}(D)$$

no new info

$$\text{FOLLOW}(\underline{L}) = \{0123456789abcdefghijklmnopqrstuvwxyz;,:)=\}$$

$$\begin{aligned} I &\rightarrow LX \\ X &\rightarrow \lambda \end{aligned} \quad \begin{aligned} I &\rightarrow L \\ X &\rightarrow LX \\ X &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} \therefore \text{FIRST}(X) &\subseteq \text{FOLLOW}(L) \\ \therefore \text{FOLLOW}(I) &\subseteq \text{FOLLOW}(L) \\ \therefore \text{FOLLOW}(X) &\subseteq \text{FOLLOW}(L) \end{aligned}$$